

Experiment 8

Reproducing an Image Using Prompts

[Submission PDF](#)

[Source folder on GitHub](#)

Prepared 2026-09-29 from the experiment materials on branch
``codex/complete-experiments``.

Source: [Ex08_Image_Reproduction_Prompts/README.md](#)

Exp-8: Reproducing an Image Using Prompts

Due: 14 Sep 2026 Reference repo: RaajaThilahar/Ex.No.8

Steps

The course reference image was not included with the supplied instructions, so this experiment uses a clearly labeled sample landscape generated for the exercise at `reference_image/sample_reference.png`.

Its objects, palette, lighting, composition and style are documented in `analysis.md`.

The exact prompts and one targeted refinement are recorded in `prompts/iterations.md`.

Two generated results are saved in `generated_images/`.

`observations.md` records a visual comparison. Scores are one-person subjective judgments from a single pair of generations, not a benchmark.

Tool: built-in image generation in Codex. The tool does not expose a repeatable seed or a model-version label in this workflow. The sample reference and both results are generated images, not a supplied course image.

Source: [Ex08_Image_Reproduction_Prompts/LAB_RECORD.md](#)

Lab Record: Exp-8: Reproducing an Image Using Prompts

Field	Entry
Name / Reg. No.	_____
Date	_____
Repo	https://github.com/RHUDHRESH/PROMPT_ENGINEERING-

Aim

Reproduce a given image through iterative prompt refinement.

Tools / Apparatus

Built-in Codex image generation; generated sample reference and two refined images

Procedure

Analysed the labeled sample landscape in `reference_image/`.

Ran a reconstruction prompt and a targeted refinement; saved both outputs in `generated_images/`.

Compared each image with the sample reference and recorded subjective scores in `observations.md`.

Observations

See `observations.md`. A fixed seed was unavailable; scores are subjective and based on one sample per version.

Result

The experiment includes a generated sample reference and two landscape recreations. The second prompt slightly improved the upper sky while keeping the main composition close.

Conclusion

Specific composition and palette language helped guide the image. Fine details still changed between runs, so the result is illustrative rather than a controlled benchmark.

Faculty signature: _____

Source: [Ex08_Image_Reproduction_Prompts/analysis.md](#)

Image analysis

The supplied course text did not include a reference image. To complete the exercise, a sample reference was generated with the built-in image generation tool and saved at `reference_image/sample_reference.png`.

Aspect	Observation
Primary subject	Layered mountain range at sunset
Secondary objects	Reflective winding river, pine trees, forest, valley mist
Colour palette	Orange and pink sky, purple ridges, dark green forest, teal river shadows
Lighting	Low sun behind the central ridge; warm backlight; calm evening mood
Composition	Wide frame; ridgeline near the upper third; river leads from the bottom-left toward the center; foreground, midground and distant ridges
Style	Photorealistic landscape photograph
Texture / detail	Fine foreground rocks and water reflections; soft atmospheric haze in distant valleys
Camera	Eye-level wide view, deep focus, landscape frame

Analysis method

- Identify the largest colour regions and the main subject.
- Locate the light source and note the direction of the shadows.
- Describe foreground, midground and background, including leading lines.
- Record the medium and style before listing small details.

Source: Ex08_Image_Reproduction_Prompts/observations.md

Observations

Version	Image	Match to reference (1-5)	What differs	Change for next version
v1	generated_images/v1.png	4	Close match in mountain layers, sunset, river, forest and mist. The sky is brighter orange than the reference and the horizon placement is not exact.	Add a pale pink upper-sky band, specify the upper-third horizon and reduce orange brightness.
v2	generated_images/v2.png	4	The upper sky is paler and the river route stays close. Cloud shapes, shoreline and mountain details still vary.	No further change; those fine details vary between generations.

The scores are visual judgments by one reviewer comparing the saved images with the sample reference. There was one generation per version, no fixed seed, and the reference itself was generated for this exercise. These scores are not a reproducible benchmark.

Conclusion

The most useful prompt elements were spatial relationships (river entering at bottom-left and leading toward the center), palette and light direction. The targeted refinement lightened the upper sky while largely preserving the composition.

Limitations: no fixed seed was available; generated details changed between samples; the image tool did not provide a quantitative similarity score.

Source: Ex08_Image_Reproduction_Prompts/prompt_checklist.md

Prompt-writing checklist (image reproduction)

- [] Subject: one clear main subject in the first words
- [] Secondary objects and their positions (left/right/foreground/background)
- [] Colours: name 3-5 dominant colours, and where they appear
- [] Lighting: source direction, time of day, quality (soft/hard), mood
- [] Composition: framing, horizon height, leading lines, depth layers
- [] Camera: angle, lens (wide/telephoto), depth of field, aspect ratio
- [] Style/medium: photo, oil painting, 3D render, anime, etc.
- [] Texture and detail level
- [] Negative constraints (where supported): text, watermark, blur, extra objects
- [] Order: most important words first (many models weight early words more)
- [] Change ONE group of details per iteration; log it
- [] Keep seed/settings fixed when comparing versions (if supported)
- [] Compare against the reference on the same 8 aspects as `analysis.md`
- [] Stop when remaining differences are random variation, not missing instructions

Source: [Ex08_Image_Reproduction_Prompts/prompts/iterations.md](#)

Prompt iterations

Tool used: built-in image generation in Codex. The tool did not expose a repeatable seed. v1 was generated with the sample reference; v2 used v1 as an image reference.

> The reference is `reference_image/sample_reference.png`. The reference itself was generated for this exercise because the course image was not included. These are single samples, not a controlled benchmark.

Reference creation prompt

Use case: reference image for a prompt-engineering lab. Asset type: photorealistic landscape photograph. Primary request: a sunset over a mountain range with a calm river winding from the bottom-left foreground toward the center peaks, and two small pine trees on the left riverbank. Composition: wide 16:9 landscape, eye-level, river as a clear leading line, layered mountain ridges, horizon on upper third. Palette: deep purple mountains, golden-orange horizon, pale pink upper sky, cool teal river shadows. Lighting: low sun hidden behind the ridge, warm rim light, thin mist between ridges. Constraints: no people, buildings, text, logos, borders, or watermark; realistic natural landscape.

v1 - Faithful recreation

Create a faithful new image inspired by the reference for an image-prompting experiment. Preserve its wide landscape composition: orange and pink sunset above layered purple mountains, central low sun behind the ridge, mist in the valley, dark pine forest, and a reflective river across the foreground. Keep the river entering from bottom-left and leading toward the center. Photorealistic, 16:9, no text or watermark.

Result: `generated_images/v1.png`.

v2 - Targeted refinement

Refine this landscape for closer composition matching. Keep the same mountain silhouettes, river route, pine trees, mist, and warm sunset. Make the horizon sit in the upper third, reduce the brightest orange glow slightly, add a pale pink band to the upper sky, and keep the reflective river entering clearly from the bottom-left. Photorealistic wide 16:9 landscape. Do not add objects, text, or watermark.

Result: `generated_images/v2.png`; v1 was supplied as the image reference.

Change log

From -> To	What changed	Why
reference -> v1	Main colors, objects, river direction, mist, wide framing	Recreate the key visible scene
v1 -> v2	Horizon placement, lighter upper sky, controlled orange brightness	Refine composition and palette while retaining the scene

Source: [Ex08_Image_Reproduction_Prompts/tool_comparison.md](#)

Tool comparison notes: DALL-E vs Stable Diffusion vs Midjourney

Only widely documented general facts; features change between versions, so check current docs.

Aspect	DALL-E (OpenAI)	Stable Diffusion (Stability AI, open weights)	Midjourney
Prompt style	Natural-language sentences; accessible via ChatGPT or API	Keyword/phrase prompts common; long sentences work less consistently on older versions	Short descriptive phrases plus parameters

Aspect	DALL-E (OpenAI)	Stable Diffusion (Stability AI, open weights)	Midjourney
Negative prompt	No dedicated negative field; describe what you want, or state exclusions in the sentence	Yes, a separate negative prompt field in most UIs (e.g. AUTOMATIC1111, ComfyUI)	--no <thing> parameter
Seed	Not user-controllable in the ChatGPT interface	Yes, user-set seed for reproducibility (with same model/sampler/settings)	--seed <n>
Aspect ratio	Chosen via size options or stated in the request	Set width/height in the UI	--ar 16:9
Other controls	Conversational edits/refinement	Sampler, steps, CFG scale, LoRAs, ControlNet, img2img, run locally	--stylize, --chaos, version flag, image prompts, upscalers
Access	Hosted service	Local or hosted; most flexible and technical	Hosted (web/Discord), subscription

Practical implications for this exercise

- Stable Diffusion gives the most reproducible comparison across v1-v4 (fixed seed, fixed settings).
- With DALL-E or Midjourney, record the full prompt and parameters and generate several variants per version.
- Record the tool, model version and date in `prompts/iterations.md`.

Appendix: Image Outputs

[Ex08_Image_Reproduction_Prompts/generated_images/v1.png](#)



[Ex08_Image_Reproduction_Prompts/generated_images/v2.png](#)



Ex08_Image_Reproduction_Prompts/reference_image/sample_reference.png

